Lie Algebras and their Representations

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Lecture 1

1 Introduction

A *Lie group* is fundamentally a group, which also a (smooth) manifold. For example, $\mathrm{GL}_n, \mathrm{SL}_n, \mathrm{SO}_n, \mathrm{Sp}_{2n}$.

> **Example 1.1**
> A prototypical example of a Lie group is the circle group S^1.

Let G be a Lie group. Then the *Lie algebra* of G is the tangent space at the identity e of G. That is,

$$\mathfrak{g} = \mathrm{T}_e G$$

$\mathfrak{g}$ is a vector space, with additional structure, which we will see later.

By taking a derivative, we turn the conjugation map

$$\begin{aligned} G &\to \mathrm{Aut}(G) \\ g &\mapsto g(\cdot)g^{-1} \end{aligned}$$

into a map

$$\mathrm{ad} : \mathfrak{g} \to \mathrm{End}(\mathfrak{g})$$

called the *adjoint*. This gives a bilinear map

$$\begin{aligned} [\cdot,\cdot] : \mathfrak{g} \times \mathfrak{g} &\to \mathfrak{g} \\ [x, y] &= \mathrm{ad}(x)(y) \end{aligned}$$

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Example 1.2

If $G = \mathrm{GL}_n(\mathbb{R})$, then we have that $\mathfrak{g} = \mathfrak{gl}_n(\mathbb{R}) = \mathrm{Mat}_n(\mathbb{R})$, with

$$[x, y] = xy - yx$$

What do Lie algebras tell us about the structure of the Lie group G?

- We will define the *root system* of $\mathfrak{g}$, and this then tells us about commutator relations in G (see Carter's book).
- We will define the *Weyl group* of $\mathfrak{g}$. For example, the Weyl group of $\mathfrak{gl}_n(\mathbb{C})$ is isomorphic to S_n, there is an embedding $S_n \hookrightarrow \mathrm{GL}_n(\mathbb{C})$, vie permutation matrices. Let B denote the Borel subgroup of upper triangular matrices in $\mathrm{GL}_n(\mathbb{C})$, then there exists a *Bruhat decomposition*

$$G = \bigsqcup_{w \in S_n} BwB$$

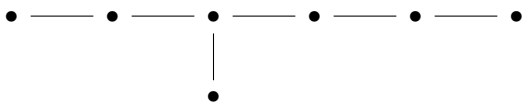
Lie algebras also give us information of the representation theory of G and $\mathfrak{g}$. For example, there exists a bijection

$$\{\text{finite dimensional } \mathbb{C}\text{-representations of } \mathrm{SL}_n(\mathbb{C})\} \leftrightarrow \{\text{finite dimensional } \mathbb{C}\text{-representations of } \mathfrak{sl}_n(\mathbb{C})\}$$

Moreover, we can describe the right hand side completely.

In addition, Lie algebras have applications in Algebraic Geometry, for example, we can use Lie algebras to build families of surfaces, of equivalenrly, algebraic curves (See book by Slodovy).

We will define the *Dynkin diagrams* of *semisimple Lie algebras*. For example,



is a Dynkin diagram of type E_7, and understanding the Dynkin diagram tells us about the singularities on the surfaces.

Moreover, Lie algebras also have applications in number theory, root systems/Weyl groups give the structure of groups over $\mathbb{Q}_p$, the p-adic integers (see paper on Moodle). Local Langlands correspondence predicts a relationship

$$\{\text{Galois theory of local fields}\} \leftrightarrow \{\text{complex Lie theory}\}$$

Finally, there any other applications, for example algebraic groups, quantum groups, theoretical physics, quantum mechanics.

2 Basic definitions and examples

Let k be a field. Most of the time, $k = \mathbb{C}$, but not always. We will sometimes point out how things can go wrong in characteristic p.

Definition 2.1 (Lie algebra)

A *Lie algebra* over k is a vector space $\mathfrak{g}$ over k, together with a bilinear pairing

$$[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$$

satisfying

1. $[xx] = 0$ for all $x \in \mathfrak{g}$,
2. the Jacobi identity
$$[x[yz]] + [y[zx]] + [z[xy]] = 0$$

Notation 2.2. Note that when clear, we will write $[xy] := [x, y]$.

Remark 2.3. In particular, we have antisymmetry, i.e.

$$[xy] = -[yx]$$

Definition 2.4 ((Lie) subalgebra)

A k-vector subspace $\mathfrak{h}$ of $\mathfrak{g}$ is a *(Lie) subalgebra* if $\mathfrak{h}$ is closed under the Lie bracket of $\mathfrak{g}$. That is, for all $x, y \in \mathfrak{h}$, $[xy] \in \mathfrak{h}$.

Example 2.5

Let V be a finite dimensional k-vector space, then

1. Let $\mathfrak{gl}(V) = \mathrm{End}(V)$, with $[xy] = xy - yx$. If we choose a basis for V, then we can identify $\mathfrak{gl}(V)$ with $\mathrm{Mat}_n(k)$. In this case, we will write $\mathfrak{gl}_n$ or $\mathfrak{gl}_n(k)$.
2. Let
$$\mathfrak{sl}(V) = \{x \in \mathfrak{gl}(V) \mid \mathrm{tr}(x) = 0\}$$
This defines a subalgebra of $\mathfrak{gl}(V)$, called the *special linear Lie algebra*, with $\dim(\mathfrak{sl}(V)) = \dim(V)^2 - 1$. The standard basis of $\mathfrak{sl}(V)$ is given by
$$E_{i,j} \text{ for } i \neq j$$
$$E_{i,i} - E_{i+1,i+1}$$
We will often write $\mathfrak{sl}_n$ for this Lie algebra.

Lecture 2

Example 2.6 (continued) 3. Suppose $\mathrm{char}(k) \neq 2$, and suppose V is endowed with a symmetric (non-degenerate?) bilinear form
$$\langle \cdot, \cdot \rangle : V \times V \to k$$
Then define
$$\mathfrak{so}(V) = \{x \in \mathfrak{gl}(V) \mid \langle xv, w \rangle + \langle v, xw \rangle = 0 \text{ for all } v, w \in V\}$$
If $M \in \mathrm{GL}(V)$ is such that $\langle v, w \rangle = v^{\mathsf{T}} M w$, so
$$\mathfrak{so}(V) = \left\{x \mid Mx + x^{\mathsf{T}} M = 0\right\}$$
We usually take
$$M = \begin{cases} \begin{pmatrix} 0 & I_\ell \\ I_\ell & 0 \end{pmatrix} & n = 2\ell \\ \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & I_\ell \\ 0 & I_\ell & 0 \end{pmatrix} & n = 2\ell + 1 \end{cases}$$
These are called the *orthogonal Lie algebra*, denoted $\mathfrak{so}_n$.

Remark 2.7. In the case $n = 2$, let

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

viewed as matrices (in $\mathfrak{sl}_2(\mathbb{C})$). Indeed, this is the standard basis of $\mathfrak{sl}_2(\mathbb{C})$. Note

$$[ef] = h$$
$$[he] = 2e$$
$$[hf] = -2f$$

We'll see that (in some sense) the structure of all semisimple Lie algebras comes from $\mathfrak{sl}_2(\mathbb{C})$.

4. Again suppose $\mathrm{char}(k) \neq 2$. Now suppose V is endowed with a non-degenerate skew-symmetric (or alternating) bilinear form $\langle \cdot, \cdot \rangle$, then

$$\mathfrak{sp}(V) = \{x \in \mathfrak{gl}(V) \mid \langle xv, w \rangle + \langle v, xw \rangle = 0\}$$

In coordinates, we take the form $\langle \cdot, \cdot \rangle$ to be the skew-symmetric form given by

$$M = \begin{pmatrix} 0 & I_\ell \\ -I_\ell & 0 \end{pmatrix}$$

where $n = 2\ell$. This is called the *symplectic Lie algebra*, denoted $\mathfrak{sp}_n$.

Exercise: Check that $\mathfrak{so}_n$ and $\mathfrak{sp}_n$ are Lie subalgebras of $\mathfrak{gl}_n$.

Example 2.8 (continued) 5. Any vector space V is a Lie algebra with $[vw] = 0$ for all v, w. We call such Lie algebras *abelian*. It is named like this, since for *linear Lie algebras*, that is, any subalgebra of $\mathfrak{gl}(V)$ where V is finite dimensional, $[xy] = xy - yx = 0$ is true if and only if x and y commute.

6. $\mathfrak{b}_n$ is the *Borel algebra* of upper triangular matrices

$$\begin{pmatrix} * & \cdots & * \\ & \ddots & \vdots \\ 0 & & * \end{pmatrix}$$

This is the Lie algebra associated to the Borel subgroup of upper triangular invertible matrices.

7. $\mathfrak{n}_n$ is the Lie algebra of strictly upper triangular matrices,

$$\begin{pmatrix} 0 & \cdots & * \\ & \ddots & \vdots \\ 0 & & 0 \end{pmatrix}$$

$\mathfrak{n}$ stands for *nilpotent*, see section 7.

3 Basic representation theory

Definition 3.1 (homomorphism, isomorphism)
A linear map $\varphi : \mathfrak{g} \to \mathfrak{h}$ between two Lie algebras is a *homomorphism* if

$$\varphi([xy]) = [\varphi(x), \varphi(y)]$$

for all $x, y \in \mathfrak{g}$. If φ is a linear isomorphism, we call φ an *isomorphism of Lie algebras*.

Definition 3.2 (representation)
A *representation* of $\mathfrak{g}$ is a Lie algebra homomorphism

$$\varphi : \mathfrak{g} \to \mathfrak{gl}(V)$$

for some vector space V.

Notation 3.3. We also call V itself a *representation*, or a $\mathfrak{g}$-module. We write $\mathfrak{g} \curvearrowright V$, and say $\mathfrak{g}$ acts on V. We will write

$$x \cdot v = xv := \varphi(x)(v)$$

The *dimension* of the representation is the dimension of V.

Example 3.4 1. Let $\dim(V) = 1$, then $\mathfrak{g} \curvearrowright V$ via $xv = 0$. This is called the *trivial representation*.

2. $\mathfrak{g}$ is a subalgebra of $\mathfrak{gl}(V)$, then the natural inclusion $\mathfrak{g} \hookrightarrow \mathfrak{gl}(V)$ is called the *defining representation*.

3. Let $x \in \mathfrak{g}$, define $\mathrm{ad}_x : \mathfrak{g} \to \mathfrak{g}$ by $\mathrm{ad}_x(y) = [xy]$. The map

$$\begin{aligned} \mathrm{ad} : \mathfrak{g} &\to \mathfrak{gl}(\mathfrak{g}) \\ x &\mapsto \mathrm{ad}_x \end{aligned}$$

is called the *adjoint representation*.

Remark 3.5. Recall e, h, f from remark 2.7, the adjoint representation of $\mathfrak{sl}_2(\mathbb{C})$ has matrices

$$\mathrm{ad}_h = \begin{pmatrix} 2 & & \\ & 0 & \\ & & -2 \end{pmatrix} \quad \mathrm{ad}_e = \begin{pmatrix} & -2 & \\ & & 1 \\ & & \end{pmatrix} \quad \mathrm{ad}_f = \begin{pmatrix} -1 & & \\ & 2 & \\ & & \end{pmatrix}$$

with respect to the basis $\{e, h, f\}$.

4. If V, W are representations of $\mathfrak{g}$, then so is their *direct sum* $V \oplus V$, via

$$x(v, w) = (xv, xw)$$

5. If V is a representation of $\mathfrak{g}$, then so is the *dual* V^*,

$$(xf)(v) = -f(xv)$$

for all $x \in \mathfrak{g}, f \in V^*, v \in V$.

6. If V, W are representations of $\mathfrak{g}$, then so is the *homomorphisms* $\mathrm{Hom}(V, W)$, via

$$(x \cdot f)(v) = x \cdot f(v) - f(x \cdot v)$$

Definition 3.6 (equivariant, isomorphism)
If V, W are representations of $\mathfrak{g}$, then a linear map $\varphi : V \to W$ is called $\mathfrak{g}$-*equivariant* for all $x \in \mathfrak{g}, v \in V$,

$$x \cdot \varphi(v) = \varphi(x \cdot v)$$

We say V, W are *isomorphic* if there exists a $\mathfrak{g}$ equivariant isomorphism (of vector spaces) $V \to W$.

Definition 3.7 (subrepresentation, irreducible)
A *subrepresentation* $W \leq V$ is a subspace $xw \in W$ for all $x \in \mathfrak{g}, w \in W$. A non-zero representation V

is *irreducible* or *simple* if the only subrepresentations of V are 0 and V.

Exercise: The trivial representation is irreducible, and so are the defining and the adjoint representation of $\mathfrak{sl}_2(\mathbb{C})$.

Definition 3.8 (completely reducible, semisimple)
A representation V is *completely reducible*, or *semisimple* if it decomposes as the direct sum of irreducible representations.

Exercise: A representation V is completely reducible if and only if for every subrepresentation W of V, there exists another subrepresentation W' such that

$$V = W \oplus W'$$

Example 3.9
If V is a representation, $W \leq V$ a subrepresentation, then the *quotient* V/W is a representation of $\mathfrak{g}$, via

$$x(v + W) = (xv) + W$$

Remark 3.10. The isomorphism theorems hold for quotient representations as well.

4 Representation theory of $\mathfrak{sl}_2(\mathbb{C})$

Recall

$$\mathfrak{sl}_2(\mathbb{C}) = \left\{ \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \;\middle|\; a, b, c \in \mathbb{C} \right\}$$

Example 4.1
We have

$$\mathfrak{b}_2 = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \right\}$$

which is the Borel subalgebra of 2×2 matrices. Let $V = \text{span}\left\{ v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$, and the defining representation of $\mathfrak{b}_2$. This is not completely reducible. To see this, set

$$V_1 = \text{span}\{v_1\}$$

and this does not have a complement.

Definition 4.2 (faithful)
A representation V of $\mathfrak{g}$ is *faithful* if the map

$$\mathfrak{g} \to \mathfrak{gl}(V)$$

is injective.

From now on, all Lie algebras and their representations will be over C, unless stated otherwise.
Let V be a representation of $\mathfrak{sl}_2(\mathbb{C})$. Recall the basis

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

of $\mathfrak{sl}_2(\mathbb{C})$. We know three representations of $\mathfrak{sl}_2(\mathbb{C})$ already.

dimension	name	action of h
1	trivial	0
2	defining	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
3	adjoint	$\begin{pmatrix} 2 & & \\ & 0 & \\ & & -2 \end{pmatrix}$

Definition 4.3 (weight space)
For $\lambda \in \mathbb{C}$, the *λ-weight space* of V is

$$V_\lambda = \{v \in V \mid hv = \lambda v\}$$

is the λ-eigenspace of h.

The following are vector space sums, not decompositions into subrepresentations.

- For the trivial representation,
$$V = V_0$$
- For the defining representation,
$$V = V_1 \oplus V_{-1}$$
- For the adjoint representation,
$$V = V_2 \oplus V_0 \oplus V_{-2}$$
where $V_2 = \langle e \rangle$, $V_0 = \langle h \rangle$, $V_{-2} = \langle f \rangle$.

For $v \in V_\lambda$,

$$h(ev) = (he)v = ([he] + eh)v = 2ev + \lambda ev = (\lambda + 2)ev$$

Hence $ev \in V_{\lambda+2}$. Similarly, $fv \in V_{\lambda-2}$.

That is, we have

$$\cdots \underset{f}{\overset{e}{\rightleftarrows}} V_{\lambda-2} \underset{f}{\overset{e}{\rightleftarrows}} V_\lambda \underset{f}{\overset{e}{\rightleftarrows}} V_{\lambda+2} \underset{f}{\overset{e}{\rightleftarrows}} \cdots$$

Definition 4.4 (highest weight vector)
A non-zero

$$v \in V_\lambda \cap \ker(e)$$

for some λ is called a *highest weight vector (of weight λ)*

Example 4.5
In the adjoint representation, e is a highest weight vector.

Lemma 4.6. Suppose $v \in V_\lambda$ is a highest weight vector. Then for all $n \geq 1$,

$$ef^n v = n(\lambda - n + 1)f^{n-1}v$$

Proof. Induction on n. For $n = 1$,

$$(ef)v = ([ef] + fe)v = (h + fe)v = \lambda v = 1(\lambda - 1 + 1)f^0 v$$

The inductive step follows similarly. □

Lemma 4.7. Suppose $v \in V_\lambda$ is a highest weight vector. Then

$$W = \text{span}\{v, fv, f^2v, \dots\}$$

is a sub-representation of V,

Proof. Suffices to show that for $w = f^n v \in W$, then $ew, fw, hw \in W$. By definition, $fw \in W$ is obvious. $ew \in W$ follows by lemma 4.6, and the $n = 0$ case is just $ew = 0$.

For hw,

$$f^n v \in V_{\lambda - 2n}$$

and so

$$hw = (\lambda - 2n)w \in W$$

□

Proposition 4.8. If V is finite dimensional, then a highest weight vector exists.

Proof. Choose any nonzero eigenvector v of h (always exists as we are working over $\mathbb{C}$). Consider

$$v, ev, e^2v, \dots$$

These are eigenvectors for h, with distinct eigenvalues. Hence the set

$$\{e^n v \mid e^n v \neq 0\}$$

is linearly independent. As V is finite dimensional, this set must be finite. Hence there must exists n such that

$$e^n v \neq 0 \quad \text{and} \quad e^{n+1} v = 0$$

Then $e^n v$ is a highest weight vector. □

Lemma 4.9. Suppose V is finite dimensional, and $v \in V_\lambda$ is a highest weight vector, then $\lambda \in \mathbb{Z}_{\geq 0}$.

Proof. Any non-zero vectors of the form $f^n v$ must be linearly independent, so there exists $n \geq 0$ such that $f^n v \neq 0, f^{n+1} v = 0$. By lemma 4.6,

$$0 = ef^{n+1} v = (n+1)(\lambda - n) f^n v$$

Hence we must have that $\lambda = n$, since $n + 1 \neq 0, f^n v \neq 0$. □

Conclusion: Suppose V is irreducible, of dimension $n + 1$. Then by proposition 4.8, a highest weight vector $v \in V_\lambda$ exists. By lemma 4.7, we have a subrepresentation

$$W = \text{span}\{v, fv, \dots\}$$

So by irreducibility,

$$\{v, fv, \dots, f^n v\}$$

is a basis, as the $f^i v$ are linearly independent, and we know from lemma 4.9 that $\lambda = n$.

Corollary 4.10. If V is an irreducible representation of $\mathfrak{sl}_2$, of dimension $n + 1$, then there exists a basis $v_0, \dots, v_n$ of V, such that the actions are:

$$hv_i = (n - 2i)v_i \qquad fv_i = \begin{cases} v_{i+1} & i + 1 \leq n \\ 0 & i = n \end{cases} \qquad ev_i = \begin{cases} i(n - i + 1)v_{i-1} & i - 1 \geq 0 \\ 0 & i = 0 \end{cases}$$

In particular, there is a unique irreducible representation of $\mathfrak{sl}_2$ with dimension $n + 1$ for all $n \geq 0$.

Remark 4.11. Let V be the $n+1$ dimensional irreducible representation of $\mathfrak{sl}_2$, and let $v \in V$ be a highest weight vector. Note

$$\left(ef + fe + \frac{1}{2}h^2\right)(v) = \left(n + \frac{n^2}{2}\right)v$$

5 Irreducible modules for $\mathfrak{sl}_2$

Notation 5.1. We will write $V(n)$ for the $n+1$ dimensional irreducible representation of $\mathfrak{sl}_2$.

Definition 5.2 (weights)
Given a representation V for $\mathfrak{sl}_2$,

$$\{\lambda \in \mathbb{C} \mid V_\lambda \neq 0\}$$

are the *weights* of V.

We will show *Weyl's theorem*.

Theorem 5.3. Every finite dimensional representation of $\mathfrak{sl}_2$ is completely reducible.

This result, along with corollary 4.10, implies that the action of h completely determines a finite dimensional representation of $\mathfrak{sl}_2$.

Example 5.4
Suppose V is a 5-dimensional representation of $\mathfrak{sl}_2$, and there exists $v \in V$ of weight 3. This means that by counting dimensions, the possible weights are $\{3, 1, -1, -3\}$, and $\{0\}$. Thus,

$$V \cong V(3) \oplus V(0)$$

We will need a few facts. Let $\mathfrak{g}$ be a Lie algebra, $\varphi : \mathfrak{g} \to \mathfrak{gl}(V)$ be a representation of $\mathfrak{g}$, and suppose there exists σ commuting with $\varphi(x)$ for all $x \in \mathfrak{g}$. Then:
Fact 1:

$$\ker(\sigma - \lambda \operatorname{id}_V)$$

is a subrepresentaion of V, for all $\lambda \in \mathbb{C}$.
Fact 2: If V is irreducible, then σ is a scalar multiple of id_V. That is, Schur's Lemma.

Definition 5.5 (Casimir element)
Let V be a finite dimensional representation of $\mathfrak{sl}_2$. Then

$$\Omega = ef + fe + \frac{1}{2}h^2 \in \mathfrak{gl}(V)$$

is called the *Casimir element (of $\mathfrak{sl}_2$)*.

In fact, Ω is *central*.

Lemma 5.6. If $\varphi : \mathfrak{sl}_2 \to \mathfrak{gl}(V)$ is a representation, then Ω commutes with $\varphi(x)$ for all $x \in \mathfrak{sl}_2$.

Proof. To show Ω is central, suffices to show $\Omega e = e\Omega, \Omega f = f\Omega, \Omega h = h\Omega$. Just compute. □

Corollary 5.7. If V is an irreducible finite dimensional representation of $\mathfrak{sl}_2$, then $\Omega \circlearrowright V$ by a scalar.

Proof. By Schur's lemma and lemma 5.6. Moreover, the scalar is

$$\frac{n^2}{2} + n$$

□

Proof of theorem 5.3. Let $\varphi : \mathfrak{sl}_2 \to \mathfrak{gl}(V)$ be a finite dimensional representation of $\mathfrak{sl}_2$. Let $W \leq V$ be a subrepresentaion. We need to find a subrepresentaion $U \leq V$, such that

$$V \cong W \oplus U$$

Case 1: W **has codimension** 1. So $V/W \cong V(0)$.

Subcase (i): W **is trivial.** In this case, $\dim(V) = 2$, and so we have a basis v_1, v_2 of V, with respect to which $\mathfrak{sl}_2$ acts on V, by matrices

$$\begin{pmatrix} 0 & * \\ 0 & 0 \end{pmatrix}$$

We will show V is isomorphic to $\underbrace{V(0)}_{=W} \oplus \underbrace{V(0)}_{=U}$. Note that

$$\left[\begin{pmatrix} 0 & x \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & y \\ 0 & 0 \end{pmatrix} \right] = 0$$

for all x, y. Since φ is a representation, it respects the Lie bracket. We must have that

$$\varphi(h) = [\varphi(e), \varphi(f)] = 0$$

and so

$$\varphi(e) = \frac{1}{2}[\varphi(h), \varphi(e)] = 0$$

and $\varphi(f) = 0$ similarly. Thus, the action of $\mathfrak{sl}_2$ is trivial.

Subcase (ii): $W = V(n)$ **is irreducible,** $n > 0$**.** Now we have $\Omega \in \mathfrak{gl}(V)$, and we will show

$$V = V(n) \oplus \ker(\Omega)$$

By Schur's lemma, and the fact that W is irreducible, and that Ω acts on V/W trivially, there is a basis for V, such that

$$\Omega = \begin{pmatrix} \lambda I & * \\ 0 & 0 \end{pmatrix}$$

W is non-trivial by assumption, and so $\ker(\Omega) \neq 0$, and $W \cap \ker(\Omega) = 0$. Hence

$$V = W \oplus \Omega$$

Subcase (iii): For a general W**.** We do this by induction on $\dim(V)$. If $\dim(V) = 1$, the result is clearly true. So we can assume $\dim(V) \geq 2$. Let $W' \leq W$ be a non-zero subrepresentation. As $\dim(W/W') < \dim(V)$, and $\operatorname{codim}_{V/W'}(W/W') = 1$, by induction, this implies that we have a splitting

$$\frac{V}{W'} = \frac{W}{W'} \oplus \frac{W''}{W'} \tag{1}$$

for some subspace $W'' \leq V$, with $W' \subseteq W''$, and W''/W' is a subrepresentation of V/W'. Moreover, W''/W' is codimension 1, and $\dim(W') \leq \dim(V)$. W'' being a subrepresentation of V follows from the fact that W''/W' is a subrepresentation. By induction again, there exists a subrepresentation $U \leq W''$ such that

$$W'' = W' \oplus U$$

We know that

$$V = W \oplus U$$

as $W \cap U = 0$ since eq. (1) is a direct sum, and so $W \cap U \leq W' \cap U = 0$. Using $\dim(U) = 1$, and counting dimensions we are done.

Case 2: Let W be arbitrary. Recall the action on $\mathrm{Hom}(V, W)$, is given by

$$(x\varphi)(v) = x\varphi(v) - \varphi(xv)$$

Define

$$\mathbb{V} = \{\psi \in \mathrm{Hom}(V, W) \mid \psi|_W = \lambda \,\mathrm{id}_W \text{ for some } \lambda\}$$

and we have a subspace

$$\mathbb{W} = \{\psi \in \mathbb{V} \mid \psi|_W = 0\} \leq \mathbb{V}$$

Since we lose one degree of freedom going from $\mathbb{V}$ to $\mathbb{W}$, and so $\mathrm{codim}_{\mathbb{V}}(\mathbb{W}) = 1$. Suppose $\psi|_W = \lambda \,\mathrm{id}_W$, $x \in \mathfrak{sl}_2, w \in W$, then

$$(x\psi)(w) = x\psi(w) - \psi(xw) = x(\lambda w) - \lambda(xw) = 0$$

So $\mathbb{V}$ is a subrepresentation of $\mathrm{Hom}(V, W)$, and so $\mathbb{W}$ is a subrepresentation as well. By case 1, there exists a one-dimensional subrepresentation $\mathbb{U} \leq \mathbb{V}$, such that

$$\mathbb{V} = \mathbb{U} \oplus \mathbb{W}$$

Write $\mathbb{U} = \langle \gamma \rangle$, for some $\gamma \in \mathbb{V}$, and so $\gamma|_W = \lambda \,\mathrm{id}_W$, for some non-zero λ.

Claim 5.8. We have a vector space decomposition:

$$V = W \oplus \ker(\gamma)$$

Lecture 5

Proof of claim. By construction, $W \cap \ker(\gamma) = 0$, and by dimension counting, $\dim(V) = \dim(W) + \dim(\ker(\gamma))$, as $W = \mathrm{im}(\gamma)$. Since dimensions add up and the intersection is zero, we have a direct sum of vector spaces. □

Finally, it remains to show that $\ker(\gamma)$ is a subrepresentation of V. Let $v \in \ker(\gamma)$, $x \in \mathfrak{sl}_2$. Since $\mathbb{U}$ is one-dimensional of $\mathfrak{sl}_2$, it must be the trivial representation. Thus,

$$0 = (x\gamma)(v) = x\gamma(v) - \gamma(xv) = -\gamma(xv)$$

as $\gamma(v) = 0$. This means that $xv \in \ker(\gamma)$ as required.

□

In Humphreys' book §6.3, Humphreys proves theorem 5.3 for a general semisimple Lie algebra. Or see Henderson §5.2.1, §7.5.1.

Remark 5.9.

1. The proof only needed (in terms of representation theory)
   - existence of a Casimir element Ω,
   - the only one-dimensional representation is the trivial representation.
2. Complete reducibility is rare, it can fail for infinite dimensional Lie algebras, or simple Lie algebras in positive characteristic. For example, the adjoint representation of $\mathfrak{sl}_n(\mathbb{F}_p)$ on $\mathfrak{gl}_n(\mathbb{F}_p)$ is not completely reducible if $p \mid n$.

6 Tensor products

Given vector spaces V, W, with bases $v_1, \ldots, v_n$ and $w_1, \ldots, w_m$ respectively. We define the *tensor product* $V \otimes_{\mathbb{C}} W$ as the $\mathbb{C}$-vector space, with basis

$$\{v_i \otimes w_j\}_{i,j}$$

subject to the usual bilinearity conditions.

Definition 6.1 (tensor product of representations)

If V, W are representations of a Lie algebra $\mathfrak{g}$, then so is $V \otimes W$, with

$$x(v \otimes w) = (xv) \otimes w + v \otimes (xw)$$

Example 6.2

If V, W are $\mathfrak{sl}_2$ representations, $v \in V_\lambda$, $w \in W_\mu$, then

$$h(v \otimes w) = (\lambda + \mu)(v \otimes w)$$

That is,

$$v \otimes w = (V \otimes W)_{\lambda+\mu}$$

Thus, the weights of $V \otimes W$ are just $\lambda + \mu$, where λ is a weight for V, μ is a weight for W.

Example 6.3

For

$$V(2) \otimes V(2) = V(2)^{\otimes 2}$$

we have the weights:

	2	0	-2
2	4	2	0
0	2	0	-2
-2	0	-2	-4

and so

$$V(2)^{\otimes 2} = V(4) \oplus V(2) \oplus V(0)$$

In particular, if v_n is a highest weight vector in $V(n)$, then $v_n \otimes v_m$ is a highest weight vector for $V(n) \otimes V(m)$.

We would like a general formula for decomposing $V(n) \otimes V(m)$. The answer, as in Part II Representation theory, is a *Clebsch-Gordon* formula

$$V(n) \otimes V(m) = \bigoplus_{r=|n-m|, r \cong n-m \pmod 2}^{n+m} V(r)$$

We won't need this though.

Definition 6.4

The *n-th symmetric power* is

$$S^n V = \mathrm{Sym}^n(V) = \frac{V^{\otimes n}}{M_n}$$

where M_n is the span of

$$u_1 \otimes \cdots \otimes u_n - u_{\sigma(1)} \otimes \cdots \otimes u_{\sigma(n)}$$

where $\sigma \in S_n$, and $u_i \in V$.

For example, M_2 is the span of $v \otimes w - w \otimes v$. In particular, note that M_n is a subrepresentation of $V^{\otimes n}$ of V. So $\mathrm{Sym}^n(V)$ is a representation of V.

Example 6.5

In $S^2 V$, $v \otimes w = w \otimes v$, and so $S^2 V$ has basis

$$v_i \otimes v_j$$

for $i \leq j$. Decomposing $S^2V(2)$, we see that $e \otimes e \in S^2V(2)$ is nonzero (note $V(2)$ is the adjoint representation), and so $V(4)$ is a subrepresentation of $S^2V(2)$. In particular, we have a splitting

$$S^2V(2) = V(4) \oplus V(0)$$

Definition 6.6

The *n-th exterior (or alternating) power* is

$$\Lambda^n V = \frac{V^{\otimes n}}{N_n}$$

where N_n is the span of

$$u_1 \otimes \cdots \otimes u_n$$

where $u_i \in V$, and $u_i = u_j$ for some $i \neq j$.

Again, N_n is a subrepresentation of $V^{\otimes n}$, and so $\Lambda^n V$ is a representation.

Example 6.7

With $n = 2$, $N_2 = \text{span}\{v \otimes v\}$.

Notation 6.8. We write (the coset of) $u_1 \otimes \cdots \otimes u_n$ in $\Lambda^n V$ as

$$u_1 \wedge \cdots \wedge u_n$$

Exercises:

1. Decompose $\Lambda^2 V(2) = V(2)$, with basis $e \wedge f, e \wedge h, h \wedge f$.
2. Find the dimensions of $S^n V$ and $\Lambda^n V$.

7 Results about semisimple Lie algebras

Let $\mathfrak{g}$ be a Lie algebra over $\mathbb{C}$.

Definition 7.1 (ideal)

A subspace $I \subseteq \mathfrak{g}$ is an *ideal* of $\mathfrak{g}$ if

$$[xy] \in I$$

for all $x \in \mathfrak{g}, y \in I$.

Remark 7.2. Any ideal is automatically a subalgebra.

Suppose I is an ideal, then $\mathfrak{g}/I$ is a Lie algebra under

$$[x + I, y + I] = [x, y] + I$$

Moreover, I is an ideal if and only if it is a subrepresentation of the adjoint representation $\text{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$.

Example 7.3

The *centre* of $\mathfrak{g}$ is

$$Z(\mathfrak{g}) = \{x \in \mathfrak{g} \mid [x, y] = 0 \text{ for all } y \in \mathfrak{g}\}$$

and by definition, $Z(\mathfrak{g}) = \ker(\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g}))$.

If $Z(\mathfrak{g}) = 0$, then $\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ is faithful, and thus we have an embedding

$$\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$$

and as such, $\mathfrak{g}$ can be regarded as a Lie subalgebra of $\mathfrak{gl}_n$, where $n = \dim(\mathfrak{g})$.

Theorem 7.4 (Ado). Suppose $\mathrm{char}(k) = 0$, then any finite dimensional Lie algebra $\mathfrak{g}$ embeds as a Lie subalgebra of $\mathfrak{gl}_m$ for some m.

Proof. Omitted. □

Note that the embedding need not be via the adjoint representation. In fact, this is true for $\mathrm{char}(k) = p > 0$, due to Iwasawa.

Example 7.5

The *derived subalgebra* of $\mathfrak{g}$ is

$$D(\mathfrak{g}) = [\mathfrak{g}, \mathfrak{g}] = \{[x, y] \mid x, y \in \mathfrak{g}\}$$

This is an ideal of $\mathfrak{g}$.

Recall $\mathfrak{sl}_n$ is a subalgebra of $\mathfrak{gl}_n$. In fact,

$$D(\mathfrak{gl}_n) = \mathfrak{sl}_n$$

Example 7.6

Suppose $\varphi : \mathfrak{g} \to \mathfrak{h}$ is a homomorphism of Lie algebras, then

$$\ker(\varphi) = \{x \in \mathfrak{g} \mid \varphi(x) = 0\}$$

is an ideal of $\mathfrak{g}$. In fact, every ideal arises in this way.

Lecture 6

Definition 7.7 (simple)

A Lie algebra $\mathfrak{g}$ is *simple* if $[\mathfrak{g}, \mathfrak{g}] \neq 0$, and the only ideals are 0 and $\mathfrak{g}$.

Example 7.8

We can show that $\mathfrak{sl}_n$ (for $n \geq 2$), $\mathfrak{so}_n$ (for $n \geq 2$) and $\mathfrak{sl}_{2\ell}$ (for $\ell \geq 1$) are simple.

Remark 7.9.

1. if $\mathfrak{g}$ is simple, then $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$.
2. if $\mathfrak{g}$ is simple, every representation of $\mathfrak{g}$ is either faithful, or the direct sum of trivial representations.
3. $\mathfrak{g}$ is simple if and only if the adjoint representation is irreducible.

Definition 7.10 (semisimple)

A Lie algebra $\mathfrak{g}$ is *semisimple* if it is the direct sum of simple ideals. That is, ideals which are simple as Lie algebras.

Example 7.11

$$\mathfrak{so}_4 \cong \mathfrak{sl}_2 \oplus \mathfrak{sl}_2$$

We will state a more 'standard' definition of semisimple Lie algebras, and show that these are equivalent.

Definition 7.12 ((lower) central series)

The *(lower) central series* of a Lie algebra $\mathfrak{g}$ is the sequence of subalgebras

$$\mathfrak{g} = \mathfrak{g}^0 \supseteq \mathfrak{g}^1 \supseteq \mathfrak{g}^2 \supseteq \cdots$$

with $\mathfrak{g}^n = [\mathfrak{g}, \mathfrak{g}^{n-1}]$. That is,

$$\mathfrak{g} \supseteq [\mathfrak{g}, \mathfrak{g}] \supseteq [\mathfrak{g}, [\mathfrak{g}, \mathfrak{g}]] \supseteq \cdots$$

Definition 7.13 (derived series, upper central series)

The *derived series*, or *upper central series* for a Lie algebra $\mathfrak{g}$ is the sequence

$$\mathfrak{g}^{(0)} \supseteq \mathfrak{g}^{(1)} \supseteq \mathfrak{g}^{(2)} \supseteq \cdots$$

where $\mathfrak{g}^{(0)} = \mathfrak{g}$, and $\mathfrak{g}^{(n)} = [\mathfrak{g}^{(n-1)}, \mathfrak{g}^{(n-1)}]$. That is,

$$\mathfrak{g} \supseteq [\mathfrak{g}, \mathfrak{g}] \supseteq [[\mathfrak{g}, \mathfrak{g}], [\mathfrak{g}, \mathfrak{g}]] \supseteq \cdots$$

Remark 7.14.

- $\mathfrak{g}^{(n)} \subseteq \mathfrak{g}^n$,
- $\mathfrak{g}^n$ and $\mathfrak{g}^{(n)}$ are ideals.

To see the second point, we induct on n. The case $n = 0$ is clear. Let $x \in \mathfrak{g}, y \in \mathfrak{g}^n$. Then

$$[x, y] \in \mathfrak{g}^{n-1}$$

since $\mathfrak{g}^{n-1}$ is an ideal, and $\mathfrak{g}^n \subseteq \mathfrak{g}^{n-1}$. As

$$\mathfrak{g}^n = \left\{ [x, y] \mid x \in \mathfrak{g}, y \in \mathfrak{g}^{n-1} \right\}$$

this clearly contains $[x, y]$.

Example 7.15

1. if $\mathfrak{g}$ is simple, then $\mathfrak{g}^n = \mathfrak{g}^{(n)} = \mathfrak{g}$ for all n.
2. if $\mathfrak{g}$ is abelian, then $\mathfrak{g}^1 = \mathfrak{g}^{(1)} = 0$.
3. let $\mathfrak{n}_n \subseteq \mathfrak{gl}_n$ be the Lie algebra of strictly upper triangular $n \times n$ matrices. The central series for $\mathfrak{n}_n$ is

$$\left\{ \begin{pmatrix} 0 & & * \\ & \ddots & \\ & & 0 \end{pmatrix} \right\} \supseteq \left\{ \begin{pmatrix} 0 & 0 & & * \\ & \ddots & \ddots & \\ & & \ddots & 0 \\ & & & 0 \end{pmatrix} \right\} \supseteq \cdots \supseteq 0$$

Note $\mathfrak{n}_3$ is the *Heisenberg Lie algebra*.

Definition 7.16 (nilpotent)

If $\mathfrak{g}^n = 0$ for some n, then $\mathfrak{g}$ is called *nilpotent*.

Definition 7.17 (solvable, soluble)
If $\mathfrak{g}^{(n)} = 0$ for some n, then $\mathfrak{g}$ is called *solvable* (or *soluble* in BrE).

Exercise: Let $\mathfrak{b}_n \subseteq \mathfrak{gl}_n$ be the Borel Lie algebra of upper triangular matrices. Then $\mathfrak{b}_n$ is solvable but not nilpotent (for $n \geq 2$).

Note on the other hand that as $\mathfrak{g}^{(n)} \subseteq \mathfrak{g}^n$, nilpotent implies solvable.

The next result allows the theory of complex semisimple Lie algebras to go 'far' with minimal work.

Theorem 7.18 (Lie's theorem). Let $k = \mathbb{C}$ (or an algebraically closed field with characteristic 0). Let $\mathfrak{g} \subseteq \mathfrak{gl}(V)$ be a Lie subalgebra, and suppose $\mathfrak{g}$ is solvable. Then there exists a common eigenvector for all elements of $\mathfrak{g}$.

That is, there exists $v \in V$ non-zero, such that for all $x \in \mathfrak{g}$, $xv = \lambda_x v$ for some $\lambda_x \in \mathbb{C}$.

Proof. Omitted. See Humphreys Theorem 4.1. □

Corollary 7.19. There exists a basis for V such that every element is upper triangular.

In fact, using theorem 7.18 and induction on $\dim(V)$, we can show that there exists a chain of subspsaces

$$0 = V_0 \leq V_1 \leq \cdots \leq V_n = V$$

with $\dim(V_i) = i$, and $\mathfrak{g}(V_i) \subseteq V_i$. By considering a basis for V_i, we get the corollary.

Using this result, we can consider $\mathfrak{g} \subseteq \mathfrak{b}_n$.

Aside: We call the sequence

$$0 = V_0 \leq V_1 \leq \cdots \leq V_n$$

a *maximal flag*, and there is interesting geometry related to this.

Proposition 7.20. Suppose I, J are ideals of $\mathfrak{g}$. Then

(i) if $\mathfrak{g}$ is solvable, then any subalgebra or quotient of $\mathfrak{g}$ is solvable.

(ii) if I is solvable, and $\mathfrak{g}/I$ is solvable, then so is $\mathfrak{g}$.

(iii) if I, J are solvable, then so is $I + J$.

Proof. (i) is clear from definitions. For (ii), choose n such that $(g/I)^{(n)} = 0$. Then this forces $\mathfrak{g}^{(n)} \subseteq I$. But then we have $\mathfrak{g}^{(n+m)} \subseteq I^{(m)}$ for each $m \geq 0$. Since we know that I is solvable, we are done.

For (iii), note that

$$\frac{I + J}{J} \cong \frac{I}{I \cap J}$$

and the right hand side is solvable, by (i), and J is solvable by assumption, and so by (ii), $I + J$ is solvable. □

Definition 7.21 (radical)
The *radical* of $\mathfrak{g}$ is $\mathrm{Rad}(\mathfrak{g})$ is the maximal solvable ideal of $\mathfrak{g}$. That is, it is the sum of all solvable ideals of $\mathfrak{g}$.

Definition 7.22 (trace form)
Suppose $\varphi : \mathfrak{g} \to \mathfrak{gl}(V)$ is a finite dimensional representation of $\mathfrak{g}$. Then the *trace form* of V (or φ) is the

symmetric bilinear form

$$(\cdot,\cdot) : \mathfrak{g} \times \mathfrak{g} \to \mathbb{C}$$
$$(x, y) = \operatorname{tr}(\varphi(x)\varphi(y))$$

Exercise: We have the invariance relation $([x, y], z) = (x, [y, z])$

Definition 7.23 (Killing form)
The *Killing form* $K(\cdot,\cdot)$ is the trace form of ad. That is,

$$K(x, y) = \operatorname{tr}(\operatorname{ad}(x)\operatorname{ad}(y)) = (x, y)_{\text{ad}}$$

Lecture 7

Theorem 7.24 (Cartan-Killing criterion)**.** For a finite dimensional Lie algebra $\mathfrak{g}$, the following are equivalent:

1. $\mathfrak{g}$ is semisimple,
2. $\operatorname{Rad}(\mathfrak{g}) = 0$,
3. the Killing form of $\mathfrak{g}$ is non-degenerate.

Remark 7.25. $\operatorname{Rad}(\mathfrak{g}/\operatorname{Rad}(\mathfrak{g})) = 0$, since a suitable ideal of $\mathfrak{g}/\operatorname{Rad}(\mathfrak{g})$ would lift to give an ideal J of $\mathfrak{g}$, containing $\operatorname{Rad}(\mathfrak{g})$, with $J/\operatorname{Rad}(g)$ solvable. Hence J is solvable, and $J \subseteq \operatorname{Rad}(\mathfrak{g})$.
Using this, $\mathfrak{g}/\operatorname{Rad}(\mathfrak{g})$ is semisimple.

Theorem 7.26 (Levi's theorem)**.** Let k be a field with $\operatorname{char}(k) = 0$, and $\mathfrak{g}$ a finite dimensional Lie algebra over k. Then there exists a Lie subalgebra $\mathfrak{g}'$ of $\mathfrak{g}$, with $\mathfrak{g}' \cap \operatorname{Rad}(\mathfrak{g}) = 0$, and as vector spaces,

$$\mathfrak{g} = \mathfrak{g}' \oplus \operatorname{Rad}(\mathfrak{g})$$

and $\mathfrak{g}'$ is isomorphic to $\operatorname{Rad}(\mathfrak{g})$, and thus semisimple. That is, the short exact sequence

$$0 \longrightarrow \operatorname{Rad}(\mathfrak{g}) \longrightarrow \mathfrak{g} \longrightarrow \mathfrak{g}/\operatorname{Rad}(\mathfrak{g}) \longrightarrow 0$$

splits. This is called the *Levi decomposition of* $\mathfrak{g}$, and $\mathfrak{g}'$ the *Levi subalgebra of* $\mathfrak{g}$.

Not proven in the course. See Fulton-Harris appendix E. □

Lemma 7.27. Let $\mathfrak{g}$ be a Lie algebra,

(i) if I is an ideal of $\mathfrak{g}$, then so is $[I, I]$,

(ii) $\operatorname{Rad}(\mathfrak{g}) = 0$ if and only if $\mathfrak{g}$ has no non-trivial abelian ideals.

Proof. For (i), if $x, y \in I, z \in \mathfrak{g}$, we need to show that

$$[z, [x, y]] \in [I, I]$$

Using the Jacobi identity,

$$[z, [x, y]] = -[x, [y, z]] - [y, [x, z]] \in [I, I]$$

as I is an ideal.

For (ii), it is clear that any abelian ideal is solvable. Conversely, if I is solvable, then the last non-zero term in the derived series of I is abelian. □

Notation 7.28. Define

$$\mathfrak{g}^\perp = \{x \in \mathfrak{g} \mid K(x, y) = 0 \text{ for all } x \in \mathfrak{g}\}$$

Lemma 7.29. $\mathfrak{g}^\perp$ is an ideal.

Proof. For $x \in \mathfrak{g}^\perp$, $y, z \in \mathfrak{g}$, then $K([x, y], z) = K(x, [y, z]) = 0$ □

Lemma 7.30. Let I be an ideal of $\mathfrak{g}$, and let K_I denote the Killing form of I. Then

$$K_I(x, y) = K(x, y)$$

for all $x, y \in I$.

Proof. Choose a basis for I, and extend it to a basis of $\mathfrak{g}$. Given $x, y \in I$, with respect to this basis,

$$\operatorname{ad}(x) = \begin{pmatrix} A & * \\ 0 & 0 \end{pmatrix}$$

where $A = (\operatorname{ad}(x))|_I$, and similarly for $\operatorname{ad}(y)$. Set $B = (\operatorname{ad}(y))|_I$. Then

$$K_I(x, y) = \operatorname{tr}(AB) = \operatorname{tr}(\operatorname{ad}(x)\operatorname{ad}(y)) = K(x, y)$$

□

Theorem 7.31 (Cartan's criterion (for solvability))**.** Suppose $\mathfrak{g}$ is a subalgebra of $\mathfrak{gl}(V)$, for V a finite dimensional vector space over $\mathbb{C}$. If $K(x, y) = 0$ for all $x \in \mathfrak{g}$, $y \in [\mathfrak{g}, \mathfrak{g}]$, then $\mathfrak{g}$ is solvable.

In other words, $\mathfrak{g}^{(1)} \leq \mathfrak{g}^\perp$.

Proof. See Humphreys 4.3 using Lie's theorem and the Jordan decomposition. For the Jordan decomposition, see §8. □

Corollary 7.32. (i) if $\mathfrak{g} = \mathfrak{g}^\perp$, then $\mathfrak{g}$ is solvable.

(ii) if $\mathfrak{g}$ is simple, then $\mathfrak{g}^\perp = 0$,

(iii) $\mathfrak{g}^\perp$ is solvable for any finite dimensional Lie algebra $\mathfrak{g}$.

Proof. (i) Consider the adjoint $\operatorname{ad} : \mathfrak{g} \to \mathfrak{gl}(V)$. The image $\operatorname{ad}(\mathfrak{g}) = \mathfrak{g}/\ker(\mathfrak{g}) = \mathfrak{g}/Z(\mathfrak{g})$. $Z(\mathfrak{g})$ is solvable, since it is abelian, and by assumption $\mathfrak{g} = \mathfrak{g}^\perp$. By theorem 7.31, $\operatorname{ad}(\mathfrak{g})$ is solvable, and so $\mathfrak{g}$ is solvable.

(ii) Since $\mathfrak{g}^\perp$ is an ideal, and $\mathfrak{g}$ is simple, $\mathfrak{g}^\perp = 0$ or $\mathfrak{g}^\perp = \mathfrak{g}$. In the second case, by (i) $\mathfrak{g}$ is solvable, contradicting the fact that $\mathfrak{g}$ is simple, as $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$.

(iii) By lemma 7.30, $(\mathfrak{g}^\perp)^\perp = \mathfrak{g}^\perp$, so by (i), $\mathfrak{g}^\perp$ is solvable. □

With this, we can now prove the Cartan-Killing criterion.

Proof of theorem 7.24. First we show (ii) $\implies$ (iii). In this case, $\mathfrak{g}$ is solvable, and so $\mathfrak{g}^\perp \leq \operatorname{Rad}(\mathfrak{g}) = 0$. Thus K is non-degenerate.

For (iii) $\implies$ (ii), let A be an abelian ideal of $\mathfrak{g}$. We will show that $A \subseteq \mathfrak{g}^\perp$. Choose $a \in A$, $y \in \mathfrak{g}$. Choose a basis for A, and extend it to a basis of $\mathfrak{g}$. That is,

$$\operatorname{ad}(a) = \begin{pmatrix} 0 & * \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad \operatorname{ad}(y) = \begin{pmatrix} * & * \\ 0 & * \end{pmatrix}$$

Computing $\operatorname{tr}(\operatorname{ad}(a)\operatorname{ad}(y))$, we find that it is zero. Thus $A = 0$. Therefore, $\operatorname{Rad}(\mathfrak{g})$ is non-zero, then $\mathfrak{g}^\perp$ is non-zero.

Now assume (ii) and (iii) hold. If $\mathfrak{g}$ is simple, then we are done. If not, choose a minimal non-zero ideal I. Let

$$\mathfrak{g}_I = \{x \in \mathfrak{g} \mid K(x, y) = 0 \text{ for all } y \in I\}$$

This is an ideal of $\mathfrak{g}$.

Claim 7.33. $\mathfrak{g} = \mathfrak{g}_I \oplus I$.

Proof. Since I is simple (by minimality), and non-abelian (by (ii)),

$$I \cap \mathfrak{g}_I \subseteq I^{\perp} = 0$$

Consider the map

$$\mathfrak{g} \xrightarrow{\sim} \mathfrak{g}^* \longrightarrow I^*$$

given by

$$x \mapsto K(x, \cdot)$$

The kernel of this map is $\mathfrak{g}_I$. Thus, $\mathfrak{g}/\mathfrak{g}_I \cong I^* \cong I$ as vector spaces. □

Repeat this argument, with $\mathfrak{g}_I$ (choosing some minimal ideal of $\mathfrak{g}_I$). We can do this as any ideal of $\mathfrak{g}_I$ is an ideal of $\mathfrak{g}$, and so $\mathrm{Rad}(\mathfrak{g}_I) = 0$.

Claim 7.34. $(\mathfrak{g}_I)^{\perp} = 0$

Proof. Since if $(\mathfrak{g}_I)^{\perp}$, then $x \in \mathfrak{g}^{\perp}$. □

This proves (i).
Finally, to show (i) $\implies$ (ii), write

$$\mathfrak{g} = \bigoplus_j I_j$$

where the I_j are simple ideals. Let $\pi_j : \mathfrak{g} \to I_j$ denote the projection. □

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